Binary MITM oracle ladder: sub-$2^{2\ell}$ holds at $\ell = 11$ and $\ell = 12$

2026-10-16. `research/binary_mitm_oracle_l11_l12_20261016/`; code `examples/binary_mitm_oracle_l8.rs` (rung arguments). **Class: engineering** — the per-target counted saving stays the 2^{ℓ}-fold design factor; the $\Theta(2^n)$ relation-collection total and the ρ frontier are unchanged.

The row this extends

The binary `decomposition` ledger row landed its first sub-$2^{2\ell}$ oracle at $\ell = 8$ (PR 1919, merge `0f10f33`) and held it through $\ell = 10$ (PR 1944, merge `58da37f`). This round completes the ladder's next target: $\ell = 11$ and $\ell = 12$, where the pair table reaches $2^{\{2\ell\}} = 2^{\{22\}}$ entries and the pairs-and-solve reference arm reaches the ceiling of its historical usability ($\ell \approx 12$).

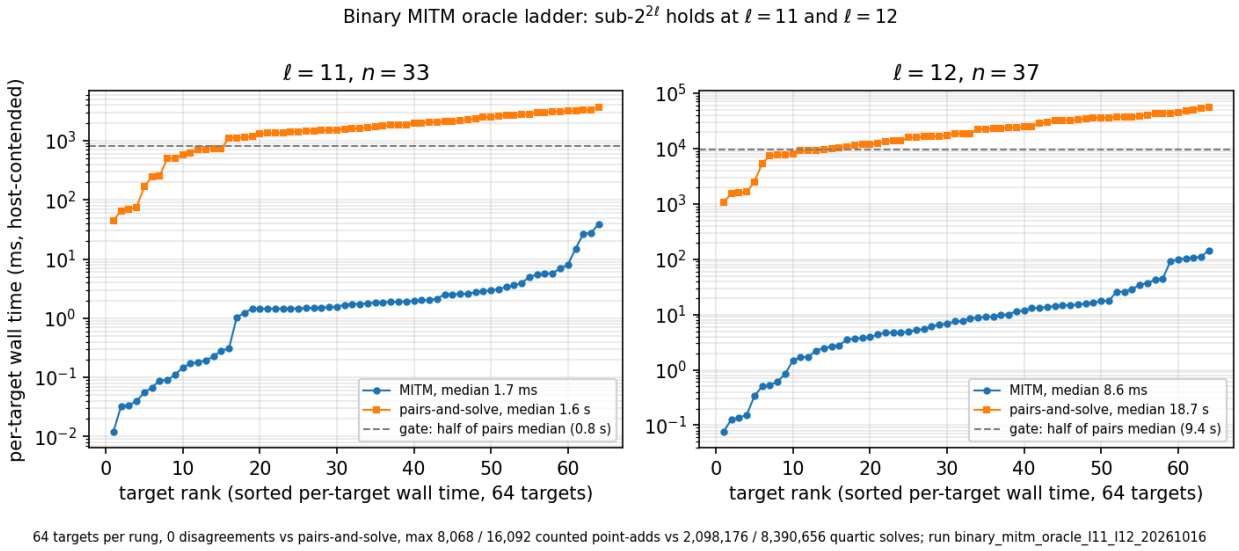
Instances (frozen per rung)

- Curve $y^2 + xy = x^3 + ax^2 + b$, $a = z \notin F_2$ (non-Koblitz), $b = 1$; deterministic target seed fixed in the example; 48 natural + 16 planted targets per rung.
- $\ell = 11$: $F_{2^{33}}$, base 2,017 points; table 2,033,876 sums, 65.1 MB, 2.69 s build.
- $\ell = 12$: $F_{2^{37}}$, base 4,023 points; table 8,092,016 sums, 258.9 MB, 14.6 s build.
- Reference arm: `SubspaceOracle::low_order(ℓ, 1)` — Gaudry pairs-and-solve on the same subspace.

Results (shared arm64 macOS host, unpinned, contended by other sessions' builds)

ℓ	n	verdicts	disagr.	max MITM ops	quartic solves	median MITM	median pairs	ratio	gate
11	33	21/43	0	8,068	2,098,176	1.746 ms	1.62 s	928.0x	PASS
12	37	24/40	0	16,092	8,390,656	8.566 ms	18.7 s	2,187.2x	PASS

$\ell = 12$ is the largest rung at which pairs-and-solve was historically usable ("usable wall-clock to $\ell = 12$" was the pre-MITM ledger state); the MITM arm decides the same 64 targets there for at most 16,092 counted point additions — the 2^{ℓ}-fold design gap against 8.4M quartic solves — with the 259 MB table charged to the factor-base stage. Every witness is re-added in the group (asserted per target); MITM verdicts agree with pairs-and-solve on all 128 targets.



Per-target wall times of both arms at both rungs (sorted, log scale) with the half-of-pairs gate lines; run `binary_mitm_oracle_l11_l12_20261016`.

Controls

- The exhaustive control ($n = 15$, $\ell = 5$, complete triple enumeration vs MITM vs pairs-and-solve, 200 targets, `cargo test --release --example binary_mitm_oracle_l8`) is unchanged and passing.
- Per-run in-code assertions: every target on-curve, every witness group-verified, every verdict cross-checked against pairs-and-solve (0 disagreements, 128/128 targets).

Claim boundary

Stage `decomposition` records only. The $2^{2\ell}$-entry table replaces the $2^{2\ell}$ per-target scan, so relation collection stays $\Theta(2^n)$: no key recovery, no end-to-end ECDLP, no `vs_rho`, no asymptotic sub-ρ claim. FFD stays `unmeasured` with reason (group-law oracle). Wall times are L0 practicality notes under external build contention; the gate metric is the paired median ratio.

Next rung

The FFD-logged algebraic control on the same systems (so the SAT/GB frontier comparison carries measured degree data), and a memory-aware $\ell = 13$ probe (table $2^{\{24\}}$ entries $\approx$ 1 GiB) if the factor-base memory gate allows.